OL P1 -MOCK

Computer Science 2210/0478

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Paper 1

1 A student is using a smartphone.

(a) Identify two input devices built into the smartphone.

1 Touchscreen

2 Microphone

[2]

(b) Identify one output device built into the smartphone.

Screen / Display / Touchscreen. **[1]**

(c) Identify one type of storage device built into the smartphone.

- SSD - **[1]**

2 Hypertext markup language (HTML) colour codes can be represented as hexadecimal.

(a) Tick (✓) **three** boxes to show which statements about the hexadecimal number system are correct.

A	It uses the values 0 to 9 and A to F.	✓
B	It can be used as a shorter representation of binary.	✓
C	It is a base 10 system.	
D	It can be used to represent error codes.	✓

(b) Denary numbers can be converted to hexadecimal.

Convert the **three** denary numbers to hexadecimal.

20 0001 0100 $(14)_{16}$

32 0010 0000 $(20)_{16}$

165 1010 0101 $(A5)_{16}$

[3]

Paper 1

3 The binary number ~~101~~00011 is stored in random access memory (RAM).

A logical left shift of **three** places is performed on the binary number.

(a) Give the 8-bit binary number that will be stored after the shift has taken place.

.... 00011000 .... **[1]**

(b) Tick (✓) **one** box to show which statement about a logical left shift of **two** places is correct.

A	It would divide the binary number by 2.	☐
B	It would multiply the binary number by 2.	☐
C	It would divide the binary number by 4.	☐
D	It would multiply the binary number by 4.	✓

8 4 2 1
2 ← 0 0 1 0
x2 4 ← 0 1 0 0
x4 8 1 0 0 0

(c) 10100011 can be stored as a two's complement integer.

Convert the two's complement integer 10100011 to denary. Show all your working.

10100011 ←
01011101
−93

1
4
8
16
64
93

−128 64 32 16 8 4 2 1
10100011
−128 (32+2+1)
−128 + 35 = −93 ✓

[2]

(d) The binary number is measured as a byte because it has 8 bits.

State how many bytes there are in a kibibyte (KiB).

.... 1024 Bytes = 2^{10} B .... **[1]**

4 A school sends student data from one computer to another using a network.

(a) What is meant by the term data packet?

In computer networks, when we send data, like a file or an email, it's usually broken down into small chunks called packets.
Each packet includes a piece of original data, destination address, source address, packet number and error checking data.

[2]

(b) The data is sent using packet switching.

Explain how packet switching works.

Packet switching is a clever method of sending data:
- split the data into packets

Each packet can take different routes through the network depending on traffic
- at the destination the packets are reassembled into the original message.

If any packets are missing or damaged, they can be re-transmitted.

[3]

(c) The data is sent using serial transmission.
Give two reasons why serial transmission is used instead of parallel transmission.

1. No signal skew.

2. More reliable over long distances. **[2]**

3. Cheaper and simpler wiring.

5 A tablet computer uses memory and storage to work with programs and data.

(a) What is the purpose of **RAM** in a computer?

RAM temporary stores programs and data that are currently in use.
It is volatile memory, meaning it loses its content when the power is switched off. **[2]**

(b) The tablet also uses **secondary storage**.

Explain why secondary storage is needed in a computer.

Secondary storage is needed to permanently store programs and files even when the computer is turned off.
It offers large storage capacity compared to RAM. **[2]**

(c) Tick (✓) **three** devices that are examples of **secondary storage**.

[ ] Random Access Memory (RAM)
[✓] Compact Disk (CD)
[✓] Blu-ray Disk
[ ] Read Only Memory (ROM)
[✓] Hard Disk Drive (HDD)
[ ] Graphics Card **[3]**

(d) The processor uses the **fetch–decode–execute (FDE) cycle** to run programs.

Describe what happens in the **cycle**.

The CPU fetches the next instruction from memory.
It decodes the instruction to identify the operation required.
It executes the instruction.
The cycle repeats for the next instruction. **[4]**

6 A computer needs firmware and system software to operate.

(a) State the purpose of firmware.

Firmware provides the essential instructions for hardware devices to operate correctly.
It is permanent software embedded into devices. **[1]**

(b) Give **one** example of firmware.

BIOS, basic input output system. Other examples are printer firmware, router firmware, UEFI. **[1]**

(c) Give **two** examples of system software.

1 Operating system, like Windows and android.

2 Utility program, like antivirus Defragmenter et cetera. **[2]**

7 A company sends secure messages using encryption.

Complete the paragraph about **asymmetric encryption**.

Choose words from the list. Some words will not be used. Use each correct word only once.

asymmetric	cipher text	encrypted key plain text		
private key	public key	decrypted	certificate	symmetric

The sender starts with some plain text

This data is encrypted using the receiver's

public key

Only the receiver can turn it back into plain text by using their private key. **[4]**

8 A factory uses an automated robot to move boxes from one conveyor belt to another.

(a) What does it mean when a robot is described as **automated**?

An automated robot can operate without human intervention, following pre-programmed instructions are responding to censor input. **[1]**

(b) List **three characteristics** of an automated robot.

1. Works without human input.

2. Uses sensors to interact with its environment.

3. Repeats tasks accurately and consistently. **[3]**

(c) The robot moves boxes and must stop when it reaches the end of the conveyor belt.

It uses **sensors** and a **microprocessor** to detect the end.

Explain how the robot uses the sensors and microprocessor to perform this task.

Sensors detect when the robot reaches the end of the conveyor belt.
Sensors send data to the micro processor.
The micro processor processes and compares the input to preset values.
Based on the input, the micro processor decides to stop the robots movement.
The micro processor sends a signal to the motor to stop the robot.

[6]

(d) Give **two** advantages of using an automated robot to move boxes in a factory.

1. Robots can work continuously without brakes, increasing productivity.

2. Robots reduce human error and performed tasks with greater accuracy.

[2]

(e) Give **two** disadvantages of using an automated robot in a factory.

1. High initial cost of purchasing and setting up robots.

2. Potential loss of jobs for human workers. It is job redundancy.

[2]

(f) The robot is upgraded to use machine learning.

Explain how this could improve the robot's performance.

The robot could learn from experience to improve task accuracy.

It could adapt to new or changing environments without reprogramming.

[2]

9 A company owner has installed a new network. Data is correct before it is transmitted across the network.

The company owner is concerned that data might have errors after transmission.

(a) Explain how the data might have errors after transmission.

Data may become corrupted due to electrical interference, signal degradation over long distances, faulty cables or connectors, network collisions are environmental noise.

[3]

(b) The company owner decides to introduce an error detection system to check the data for errors after transmission.
The error detection system uses an odd parity check and a positive automatic repeat query (ARQ).

(i) Describe how the error detection system operates to check for errors.

A parity bit is added to each bite to ensure the total number of ones is odd.
The receiver checks the parity of received data.
If parity is correct, NACK is sent back. ACK is equal to positive acknowledgement.
If parity is incorrect, a NACK is sent, or no response triggers a timeout. NACK is equal to negative acknowledgement.

This repeats until correct data is acknowledged or retries are exhausted.

[8]

(ii) Give **two** other error detection methods that could be used.

1 Checksum

2 Echo Check **[2]**

(c) The company owner also installs a firewall to help protect the network from hackers and malware.

(i) Explain how the firewall operates to help protect the network.

A firewall monitors incoming and outgoing network traffic.
It blocks unauthorised access based on IP addresses, protocols, ports, and predefined rules.
It inspects packets contents for suspicious patterns.
It alerts administrators about suspicious activities.
It acts as a barrier between trusted and unrusted networks. A hacker cannot gain access because of the firewall to the internal trusted network.

[5]

(ii) Give **two** examples of malware that the firewall can help protect the network from and how the malware work.

1 Virus: attaches to the file and spreads when opened, damaging or corrupting files and systems.

2 Worm: self replicate across networks without human actions, consuming bandwidth and crashing systems.

[4]